

Word Decoding in Children with Non-Syndromic Cleft Lip and/or Palate: Meta-Analysis

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Purpose

Past research has identified word decoding as an area of concern for children with NSCL/P (Collett et al., 2009). For example, Collett and colleagues (2009) found that differences in word decoding skills between children with and without NSCL/P continues to grow as the difficulty in reading increases.

The research examining word decoding skills in children with NSCL/P has a number of inconsistent reports, with some studies reporting word decoding skills below that of non-cleft peers (Chapman, 2011) while other studies have found no difference (Collett, Leroux, & Speltz, 2010).

Research Question:

Do children with NSCL/P demonstrate lower word decoding skills compared to their non-cleft peers?

Methods

Search Methods: Pubmed and PsychINFO were searched for articles. The inclusion terms cleft, read*, and child or adolescent. The exclusion terms were autism and surgery or surgical. A full review was completed to identify articles that met inclusion criteria. Means and standard deviations were calculated to find Cohen’s d (Table 2). Demographic information was pulled from each study that qualified (Table 3).

Data extraction & Reliability: A trained undergraduate reviewed articles for inclusion and extracted effect sizes. A graduate or postdoctoral associate reviewed 100% of entries for reliability.

Methods

Search Results	Duplicates Removed	Title/Abstract Review	Excluded	Qualified for Full Review	Qualified for Data Extraction	Effect Sizes Extracted
191	9	182	138	44	11	49

Table 1. Review of search results for inclusion.

Statistical analysis:

- Effect size d was transformed to effect size g to correct for small sample size
- Robust variance estimation was used to meta-analyze effect sizes
- Utilized REDCap (data management) and R (data analysis)

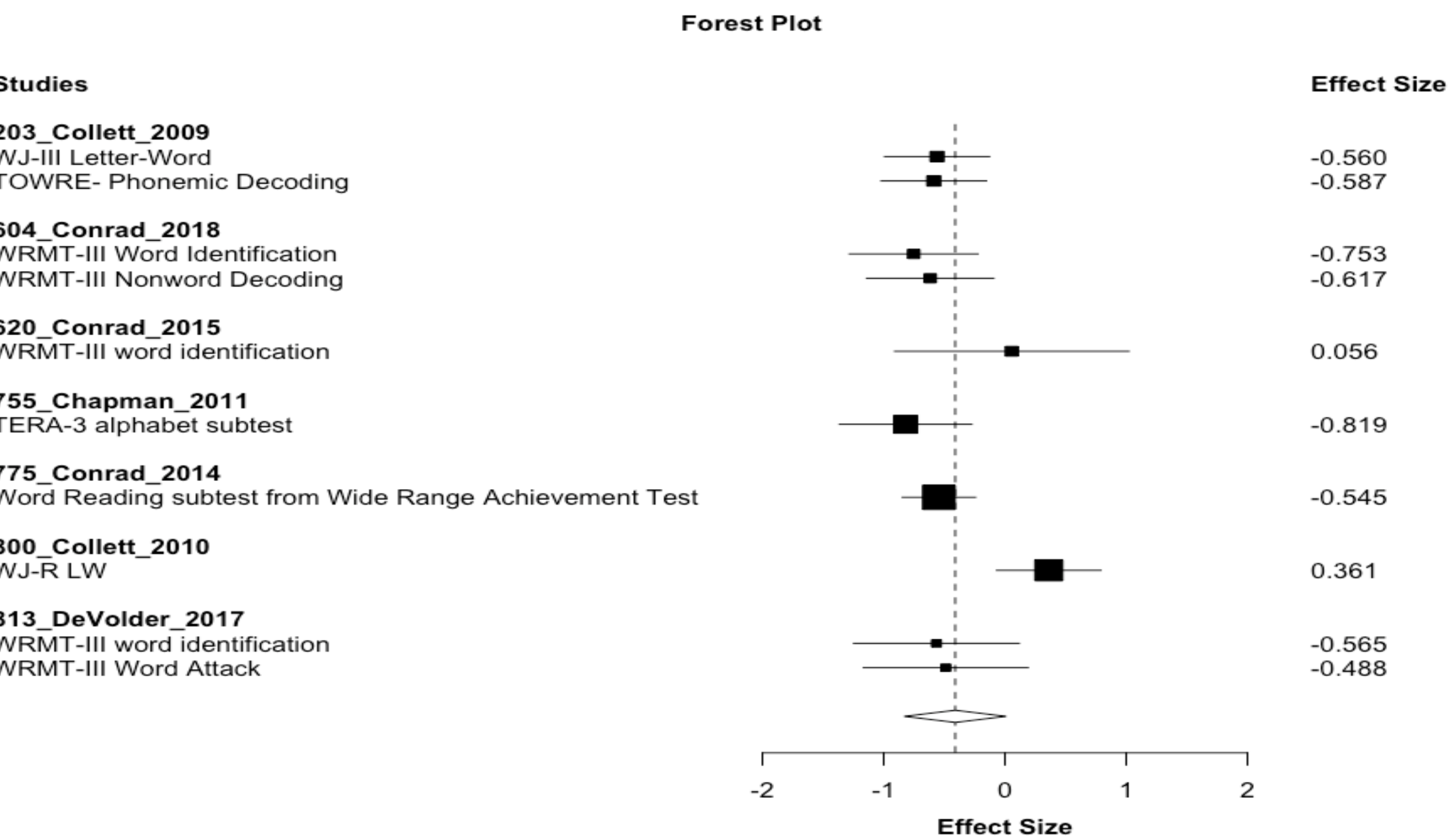
Inclusion Criteria:

- Age range of 3 to 18 years
- Included NSCL/P and typically developing groups
- At least one measure of reading, pre-reading, or reading correlate skill

Exclusion Criteria:

- Included participants with syndromes

Results



Forest Plot 1. Effect sizes calculated from means and standard deviations.

Conclusion

- Children with NSCL/P are at an increased risk for developing a reading difficulty, especially in their skills for word decoding.
- Reading, however, is largely understudied in comparison to the speech and language abilities for children with NSCL/P.
- Through this meta-analysis, we have found initial evidence demonstrating word decoding abilities in children with NSCL/P to be less than their typically developing peers.
- This study may provide important knowledge for educators and service-providers who closely monitor skill development and offer timely intervention for children with NSCL/P who may be at risk for reading difficulties

	n	k	I sq.	Tau sq.	ES _g	SE	t	df	p	95% CI
Word Decoding	7	10	66.39	0.12	-0.41	0.169	-2.43	5.6	0.054	-0.83, 0.01

Table 2. Meta-Analysis Results: Mean effect size and confidence interval reported.

	NSCL/P Means (SD)	Non-Cleft Means (SD)
Percent Male	76.2 (23.13)	75.77 (23.69)
Mean Age	118.8 (49.19)	119.8 (49.81)
Sample Size	39.14 (28.57)	41.88 (28.02)

Table 3. Demographic Results.

For More Information

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